Section 1. Multiple Choice (1 pt each)

1. What is the approximate speed of light in a vacuum?
   - (a) $3.0 \cdot 10^7$ m/s
   - $\boxed{\text{(b)}}$ $3.0 \cdot 10^8$ m/s
   - (c) $3.0 \cdot 10^9$ m/s
   - (d) $3.0 \cdot 10^{10}$ m/s
2. What is the wavelength of a photon with a frequency of 533 terahertz.
   - $\boxed{\text{(a)}}$ $5.62 \cdot 10^{-7}$ m
   - (b) $1.60 \cdot 10^{23}$ m
   - (c) $1.78 \cdot 10^6$ m
   - (d) $7.36 \cdot 10^{-9}$ m
3. What region of the electromagnetic spectrum is a photon with energy $1.18 \cdot 10^{-17}$ joules categorized in?
   - (a) Microwave
   - (b) Infrared
   - (c) Visible
   - $\boxed{\text{(d)}}$ Ultraviolet
4. As a wave travels through space, a stationary observer records a passing crest every 10 seconds. What is the frequency of this wave in hertz?
   - (a) 10 Hz
   - (b) 5 Hz
   - $\boxed{\text{(c)}}$ 0.1 Hz
   - (d) 0.2 Hz
5. What is the resistive component of the feed-point impedance of an ideal half-wave dipole antenna?
   - (a) 50Ω
   - (b) 43Ω
   - $\boxed{\text{(c)}}$ 73Ω
   - (d) 52Ω
6. What is the range of the visible portion of the electromagnetic spectrum?
   - $\boxed{\text{(a)}}$ 400 to 700 nm
   - (b) 400 to 700 μm
   - (c) 40 to 70 nm
   - (d) 40 to 70 μm

7. What region of the atmosphere reflects radio waves?
   (a) Troposphere
   (b) Mesosphere
   (c) Stratosphere
   $\boxed{\text{(d)}}$ Ionosphere
8. What is the energy of a photon with wavelength 22 nm?
   (a) $1.11 \cdot 10^{17}$ J
   (b) $4.37 \cdot 10^{-33}$ J
   (c) $1.00 \cdot 10^{-34}$ J
   $\boxed{\text{(d)}}$ $9.03 \cdot 10^{-18}$ J
9. An electromagnetic wave is composed of an oscillating electric and magnetic field. What orientation is this magnetic field to the electric field in an EM wave?
   $\boxed{\text{(a)}}$ In-Phase, Perpendicular
   (b) In-Phase, Parallel
   (c) Out-of-Phase, Perpendicular
   (d) Out-of-Phase, Parallel
10. Which of the following is a common radio band used by Wifi?
   (a) 20 THz
   (b) 5.8 THz
   $\boxed{\text{(c)}}$ 2.4 GHz
   (d) 15 MHz
11. How does the speed of light in air compare to the speed of light in a vacuum?
   $\boxed{\text{(a)}}$ Light travels slower in air
   (b) Light travels faster in air
   (c) Light travels the same speed in air
   (d) Light can travel both faster and slower in air
12. Imagine a photon with a frequency of 300 THz and a photon with a frequency of 300 MHz. If both are traveling in a vaccuum, which is traveling faster?
   $\boxed{\text{(a)}}$ Both are traveling at the same speed
   (b) The 300 THz photon
   (c) The 300 MHz photon
   (d) It is impossible to tell

13. A particular wave has a wavelength of 24 m. If a stationary observer observes the crest of that wave passing every 8.0 seconds, what is the velocity of the wave?

    (a) 0.33 m/s

    (b) 0.75 m/s

    (c) 1.5 m/s

    $\boxed{\text{(d)}}$ 3.0 m/s

14. Which of the following is used to indicate the speed of information transfer across a radio connection?

    (a) Frequency

    $\boxed{\text{(b)}}$ Bauds

    (c) Signal Bandwidth

    (d) Signal-to-Noise Ratio

15. How would the directivity of a parabolic antenna compare to a dipole antenna?

    (a) Both probably have the same directivity

    (b) The dipole has a great directivity

    $\boxed{\text{(c)}}$ The dipole has a lower directivity

16. If a half-wave dipole is transmitting in the 2.0 m band, approximately how long is the dipole?

    (a) 8.0 m

    (b) 4.0 m

    (c) 2.0 m

    $\boxed{\text{(d)}}$ 1.0 m

17. Choose the correct ordering of frequencies from highest to lowest.

    (a) infrared, microwaves, gamma rays

    (b) gamma rays, microwaves, infrared

    $\boxed{\text{(c)}}$ gamma rays, infrared, microwaves

    (d) microwaves, infrared, gamma rays

18. In a Yagi-Uda antenna, which of the following number of directors will result in the highest maximum gain?

    $\boxed{\text{(a)}}$ 8 Directors

    (b) 6 Directors

    (c) 4 Directors

    (d) 2 Directors

19. What is the relationship between velocity and frequency in regards to radio waves.
    (a) No Relationship
    (b) Constant Relationship
    (c) Direct Relationship
    $\boxed{\text{(d)}}$ Inverse Relationship
20. What is the rearmost element of a Yagi-Uda antenna called (the one furthest from the direction of maximum gain)
    (a) Driven Element
    $\boxed{\text{(b)}}$ Reflector
    (c) Director
    (d) Dipole

Section 2. Matching (2 pts each)

(x) Monopole Antenna

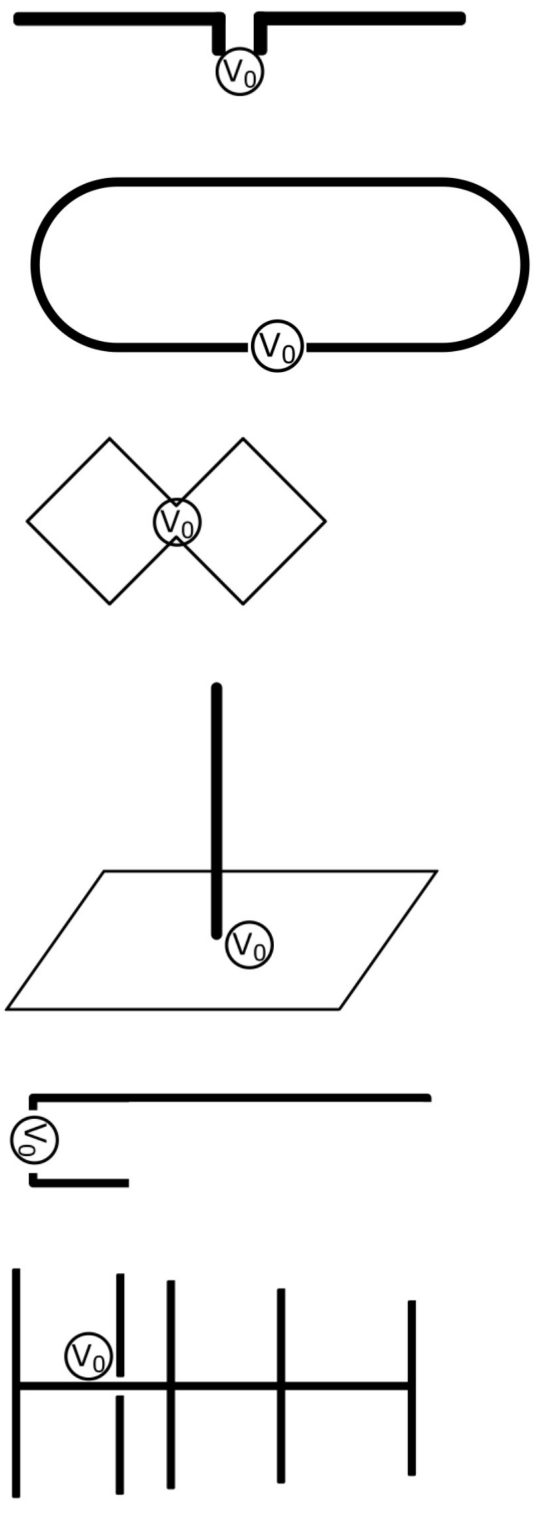
(w) Bi-Quad Antenna

(y) J Antenna

(v) Folded Dipole Antenna

(z) Yagi-Uda Antenna

(u) Dipole Antenna



Section 3. Short Answer (5 pts each)

21. List the following antenna types from greatest directivity to least directivity: dipole, isotropic, axial-mode helical.

    Answer: Axial-mode helical, dipole, isotropic (5pts)

22. Elmer's half-wave dipole antenna has a directivity of 1.64 and an efficiency of 23.0%. If Elmer wants to radiate 50.0 W of power from his antenna, how much power should he send to his antenna?

    Answer: 2 pts for Answer, 3pts for Work

$$\text{Efficiency} = \frac{P_{\text{radiated}}}{P_{\text{input}}}$$

$$P_{\text{input}} = \frac{P_{\text{radiated}}}{\text{Efficiency}}$$

$$P_{\text{input}} = \frac{50.0 \text{ W}}{0.230}$$

$$P_{\text{input}} = 217 \text{ W}$$

23. Define the concept of bandwidth. When using the minimum required bandwidth, would a morse code transmission or an FM voice transmission require more bandwidth and why?

    Answer: Bandwidth is the range of frequency occupied by an emitted radio signal (3 pts). An FM voice transmission requires more bandwidth since it must send more information in the same period of time (e.g. pitch and intensity of a voice versus the on or off signal of morse) (2 pts).

24. When transmitting from his amateur radio station, Elmer always ensures the lowest possible amount of spurious (i.e. unwanted) emissions from his antenna. His rule of thumb is an emitted power of spurious emissions 50.0 dB less than his desired signal. If the emitted power of his desired signal is 100. W, what is the maximum power he can allow for spurious emissions?

    Answer: 2 pts for Answer, 3 pts for Work

$$\text{dB} = 10 \cdot \log\left(\frac{P_{\text{emitted}}}{P_{\text{spurious}}}\right)$$

$$10^{\frac{\text{dB}}{10}} = \frac{P_{\text{emitted}}}{P_{\text{spurious}}}$$

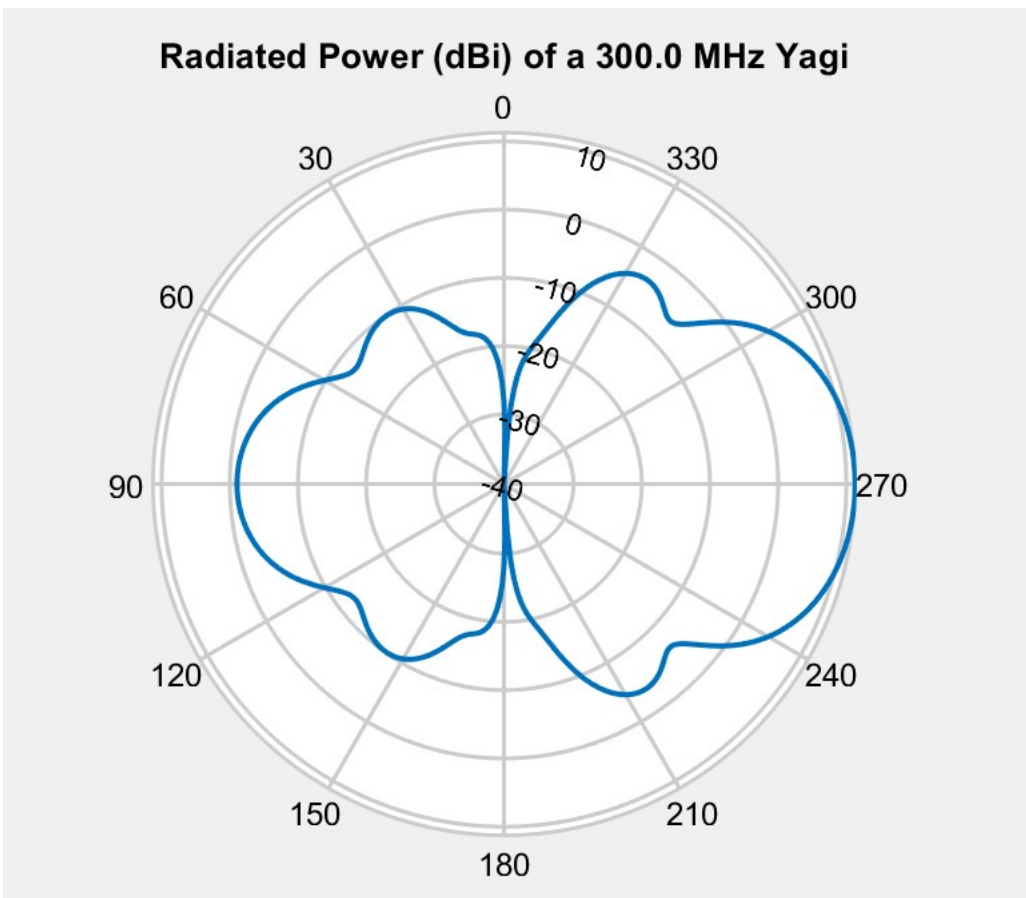
$$10^{\square} = \frac{P_{\text{emitted}}}{P_{\text{spurious}}}$$

$$\frac{P_{\text{emitted}}}{10^5} = P_{\text{spurious}}$$

$$\frac{100\ \text{W}}{10^5} = P_{\text{spurious}}$$

$$P_{\text{spurious}} = 1\ \text{mV}$$

25. (*) Below is the far-field radiation pattern for a real Yagi-Uda antenna. Answer the following questions about this radiation pattern. (This question is worth **10 pts**, not 5 pts).



(a) What is the half-power beamwidth of this antenna?

(b) What is the front to back ratio of this antenna in dB?

(c) In which direction is the peak power radiation of this antenna?

(d) If a plot of the radiation pattern of a 300 MHz isotropic antenna were added to this same plot, where would the two plots intersect? How do you know this?

Answer: (a) 44 Degrees. The Half-Power beamwidth is the angle that includes the peak power and encompasses the area that is less than 3dB less than peak power (2 pts).

(b) 12.35 dB. The ratio is just the difference of the power in dB emitted from the front and the power in dB emitted from the back (2 pts).

(c) The direction of peak radiation is 270 degrees (2 pts).

(d) The plots would intersect at the angles where the power emitted by the Yagi is 0 dBi (2 pts). This is because dBi is defined as the power emitted in comparison to an isotropic antenna. This means 0 dBi is the power emitted by an isotropic antenna (2 pts).

26. What is the concept of "reciprocity" in regards to antennas?

Answer: Reciprocity is the idea an antenna will exhibit identical behavior whether it is transmitting or receiving (5 pts). For example, if the antenna exhibits a certain gain in a certain direction when receiving, it will exhibit the same gain when transmitting.

27. (*) Elmer is setting up a Wifi router in his home, which is located in the center of his house. He bought some antennas from the store, which include a 2.4 GHz broadside helical antenna (one where the length is much longer than the diameter), a 2.4 GHz log-periodic and a 0.8 GHz dipole antenna. Which is the best option for Elmer to get coverage over his entire house and why?

Answer: The helical antenna is the best choice (2 pts). The log-periodic antenna is a directional antenna, meaning he will not get uniform coverage around his entire home (1 pt). The dipole antenna is not designed to transmit in the radio bands used by wifi, which are 2.4 GHz and 5.8 GHz (1 pt). The helical antenna when sufficiently long has a radiation pattern similar to a dipole, making it a good choice for uniform coverage around his house (1 pt).

28. (*) Elmer is interested in setting up a satellite communication system. He currently has a transceiver and a motorized antenna mount (a mount that can change an antenna's orientation). He now needs to decide whether to buy a yagi-uda, dipole, or monopole antenna for his system. Which is the best choice and why?

Answer: Since the satellite is very far from Elmer, it would be best to concentrate as much of the signal as possible towards the satellite (2 pts). The directional antenna, the yagi-uda, is the best for this application (3 pts). Since Elmer already has a motorized mount, he doesn't need to worry about the movement of the satellite across the sky.

29. Define "gain" as it relates to antennas. (Be sure to include enough information in your answer to differentiate it from similar antenna concepts).

Answer: Gain is the ratio of power transmitted by an antenna in a certain direction to the power transmitted by a loss-less isotropic antenna (3 pts). Unlike directivity, gain takes into account the efficiency of the antenna (2 pts).

30. What is the Radar Equation generally used to calculate? Write down the radar equation and define all its variables.

Answer: The Radar Equation is used to calculate the power received by a radar setup after a radar pulse is reflected off an object a certain distance away (1 pt). Each 0.5 pts

$$P_{\square} = \frac{P_{\square} G_{\square} G_{\square}^2 \lambda^2 \sigma}{(4\pi)^3 R_{\square}^2 R_{\square}^2}$$

$P_{\square} =$ Power received by the antenna
$G_{\square} =$ The gain of the transmitting antenna
$G_{\square} =$ The gain of the receiving antenna
$\lambda =$ The Transmitted wavelength
$\sigma =$ The "radar cross section" of the target
$R_{\square} =$ The distance of the transmitter to the object
$R_{\square} =$ The distance of the receiver to the object

31. What is Impedance? How much impedance does a 5.0Ω resistor have when a 60Hz AC current is applied?

Answer: Impedance is the measure of resistance to an AC Current (3 pts). The impedance of the resistor is 5.0Ω since the imaginary part of the impedance is zero (2 pts).